



Stony Brook Institute
at Anhui University

安徽大学纽约石溪学院

ISE 333: User Interface Development (Design)

Chapter 8: Navigation, Form Fill-In, Dialog Boxes

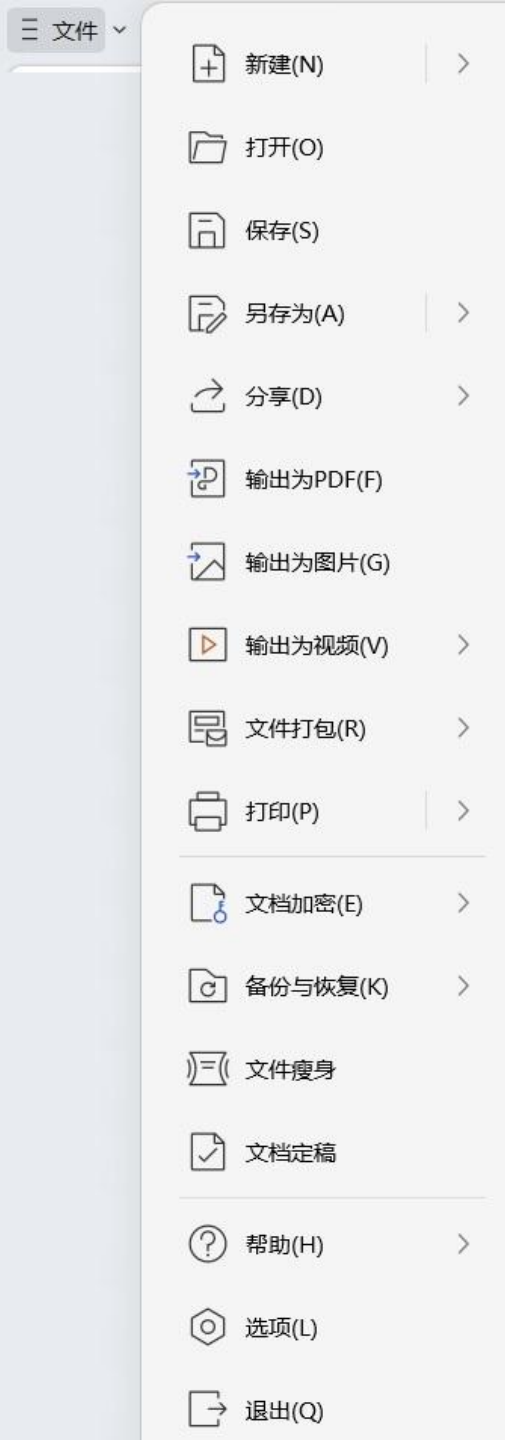
Spring 2026

Instructor: Hao Li

Navigation By Selection

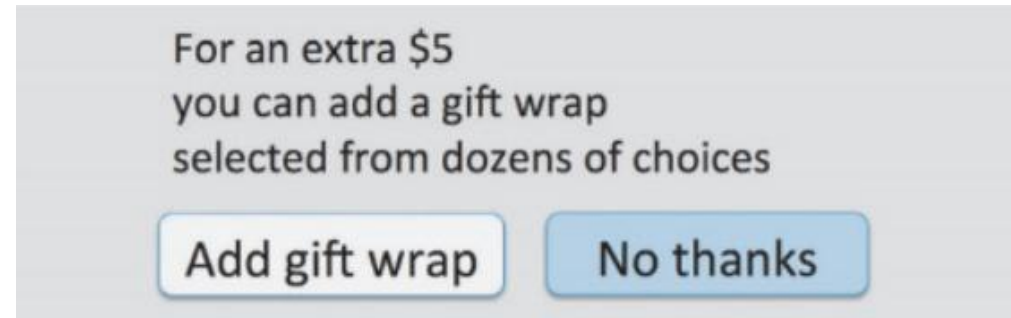
- Conception

- Navigation harnesses users' ability to rapidly skim choices, recognize what is relevant, and select what they need to realize their intentions
- The goal for UIDers is to enable fluid navigation that allows users to gracefully and confidently get to where they want to go, explore novel possible routes, and back-track when necessary
- Choices can be presented explicitly, in that there is an orderly enumeration of the items with little extraneous information
- Choices can also be embedded in text or graphics and still be selectable



Navigation By Selection

- Binary Menu
 - The simplest case of explicit menus
 - Typical choices
 - Yes/No
 - True/False
 - Previous/Next
 - Agree/Disagree
- Grid Menu
 - A small set of icons and labels



Navigation By Selection

- Radio Buttons
 - Support single-item selection from a multiple-item menu
- Check Boxes
 - Support selection of one or more items in a menu

Does anyone in your household currently smoke?

☐ Yes, someone does

☒ No, no one does

☐ Not sure

What treatment would you like to learn about?

☐ Surgery

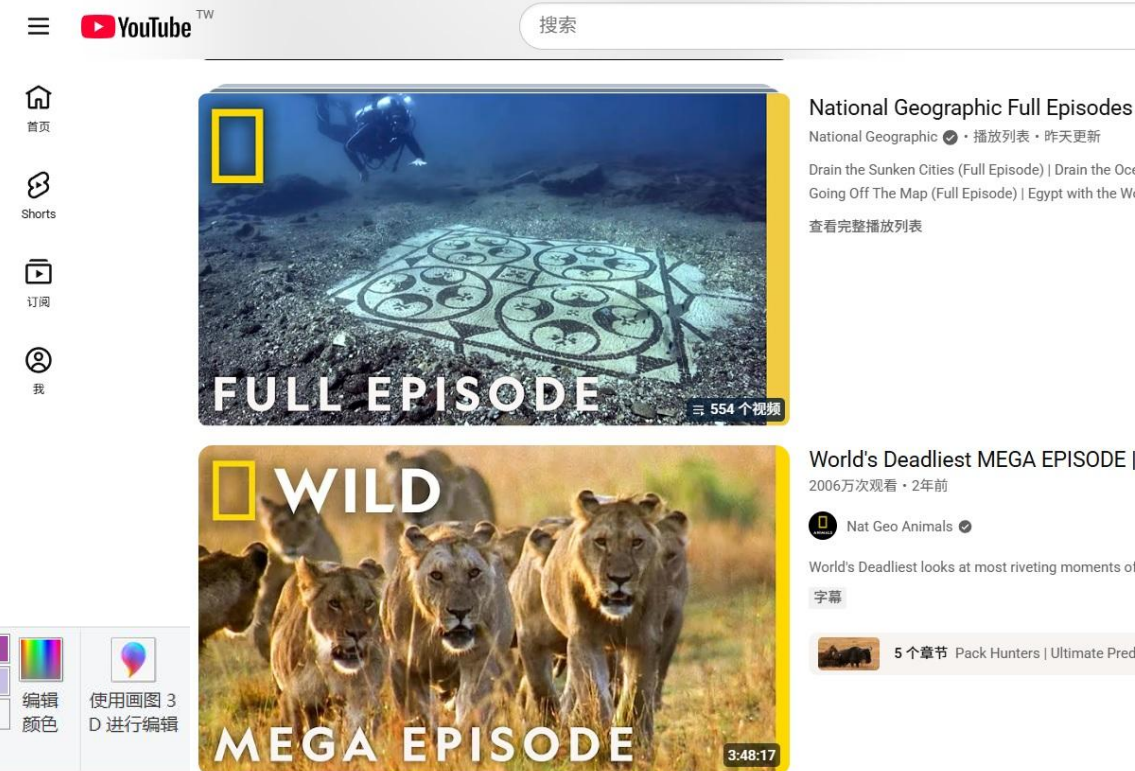
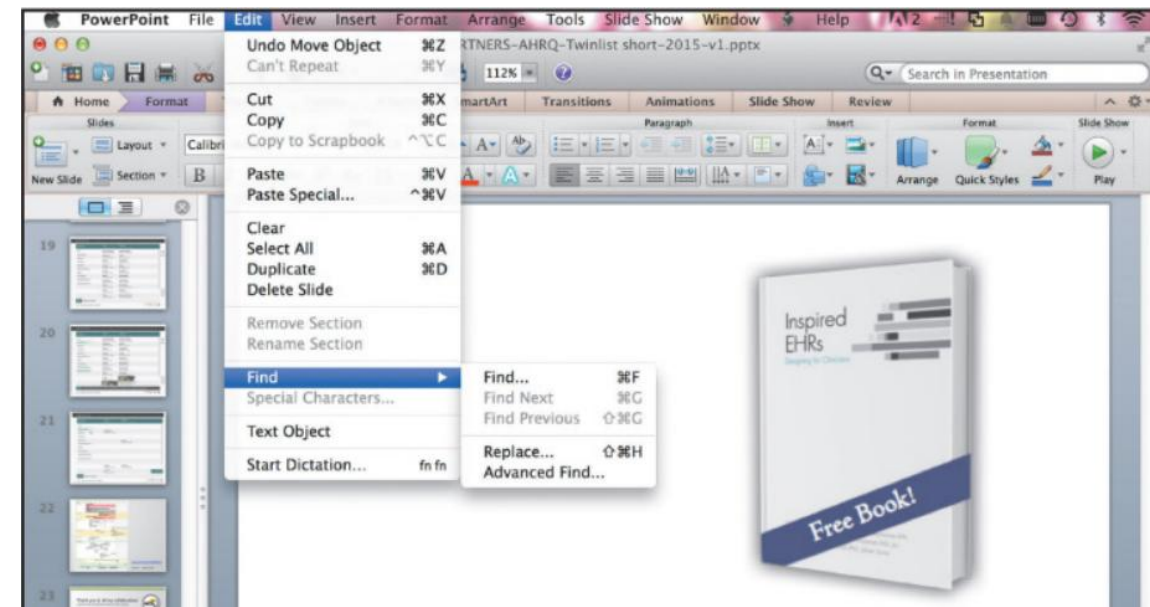
☒ Physical therapy

☒ Medication

☐ Acupuncture (not available in your plan)

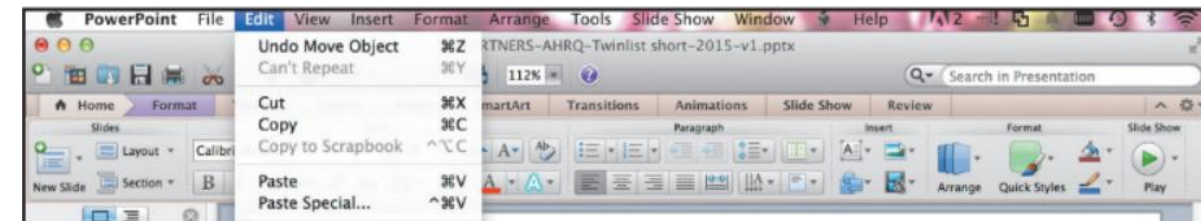
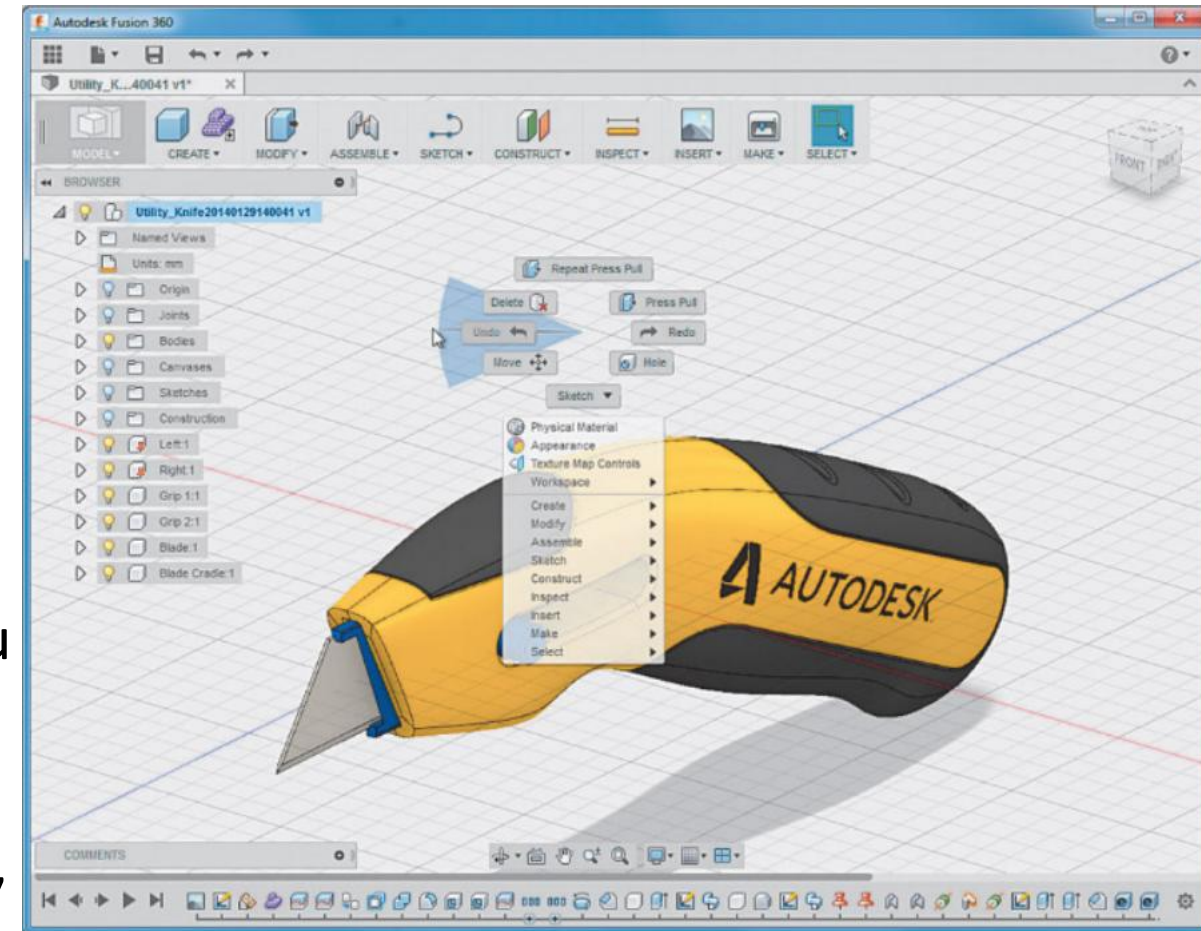
Navigation By Selection

- Menu Bar
 - Typically found at the top
- Pull-Down Menu
 - ► → Sub-menu
 - ... → Dialog box
- Hamburger Menu 
 - Access a separate menu screen
- Toolbar
- Palette



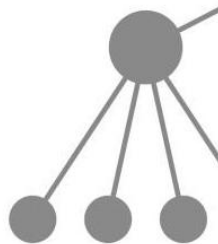
Navigation By Selection

- Pop-Up Menu
 - Appears in response to a click or tap with a pointing device
- Context Menu
 - When the content of the pop-up menu depends on the cursor position
- Ribbons
 - Introduced by Microsoft in Office 2007
 - Tabs grouping commands by task
- Scrolling Menu
 - For a long list



Content Organization

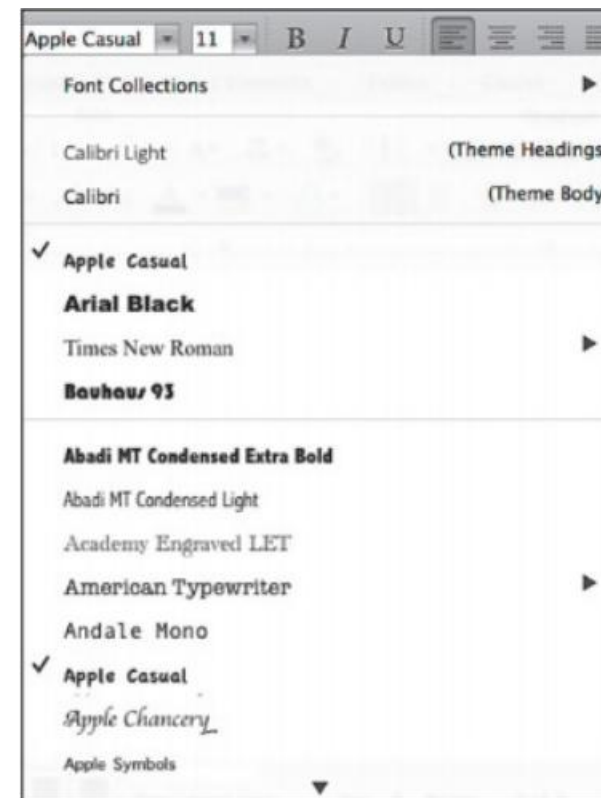
- Tree Structure
 - Logically clear and appealing for usage
- Network Structure
 - Sometimes network structures are more appropriate



- Use task semantics to organize menus.
- Limit the number of levels (i.e., prefer broad–shallow to narrow–deep).
- Create groups of logically similar items: e.g., Level 1: countries, Level 2: states, Level 3: cities.
- Form groups that cover all possibilities: e.g., age ranges: [0–9] [10–19] [20–29] and [≥ 30].
- Make sure that items are non-overlapping: e.g., use “Concerts” and “Sports” over “Entertainment” and “Events”.
- Arrange items in each branch by natural sequence (not alphabetically) or group related items.
- Keep ordering of items fixed (or possibly duplicate frequent items in dedicated sections of the menu).

Content Organization

- Presentation Sequence
 - Using frequency of use does speed up selection of the topmost items, but the loss of a meaningful ordering for low-frequency items may be disruptive
 - Adaptive split-menu strategy



Content Organization

- Presentation Sequence
- Phrasing
 - Use familiar and consistent terminology
 - Ensure that items are distinct from one another
 - Use concise phrasing
 - Bring the keyword forward
- Layout Consistency
 - Titles
 - Items
 - Instructions
 - Error messages

Form Fill-In

- Elements of Form Fill-In UID
 - Meaningful title
 - Comprehensive instructions
 - Label the fields
 - Place labels in consistent locations
 - Limit data entry
 - Make sure all fields are really needed
 - Explanatory messages for fields
 - Error prevention
 - Users cannot enter incorrect values
 - Error recovery
 - Summarize errors at the page top

The screenshot displays the 'Create an IEEE Account' form. At the top, the title 'Create an IEEE Account' is followed by a help icon. The form is divided into two main sections: 'Provide your personal information' and 'Enter e-mail address & password'.

Provide your personal information:

- * Required field**
- * Given/First name:** Input field with 'Catherine' entered.
- Middle name:** Input field.
- * Last/Family/Surname:** Input field with 'Plaisant' entered.

Enter e-mail address & password:

- The e-mail provided here will be the username of your account.
- * E-mail address:** Input field with 'cplaisant@' entered. A red border indicates an error.
- * Re-enter e-mail address:** Input field.
- * Password:** Input field with '*****' entered.
- * Confirm password:** Input field.

Feedback messages:

- A yellow warning box states: 'The e-mail address provided is not in a valid e-mail format (for example: j.doe@nomail.com). Please try again.'
- A green success box states: 'Your password is good'. Below it, a note says: 'Passwords must be between 8 and 64 characters, and include at least one number. [More...](#)'

Set security questions:

- For your security, IEEE Accounts are required to have two security questions and answers.
- * Security question 1:** A dropdown menu with 'Select' chosen.
- * Type your answer:** Input field.
- * Security question 2:** A dropdown menu with 'Select' chosen.
- * Type your answer:** Input field.

Footer:

- [Privacy & Opting Out of Cookies](#)
- [Create Account and Continue Joining](#)
- [Cancel](#)

Form Fill-In

- Elements of Form Fill-In UID
 - Immediate feedback
 - Immediate feedback about errors is preferable
 - Logical grouping
 - Related fields should be adjacent and should be aligned with blank spaces for separation between groups
 - Sequencing of fields
 - The sequencing should reflect common patterns
 - Visually appealing layout
 - Alignment creates a feeling of order and comprehensibility

The screenshot displays the 'Create an IEEE Account' form, which is organized into several sections to illustrate effective form design principles:

- Sectioning and Grouping:** The form is divided into distinct sections: 'Provide your personal information', 'Enter e-mail address & password', 'Set security questions', and a footer with links for 'Privacy & Opting Out of Cookies' and 'Create Account and Continue Joining'.
- Logical Grouping:** Related fields are grouped together. For example, 'Given/First name', 'Middle name', and 'Last/Family/Surname' are grouped under 'Provide your personal information'. Similarly, 'E-mail address', 'Re-enter e-mail address', 'Password', and 'Confirm password' are grouped under 'Enter e-mail address & password'.
- Immediate Feedback:** The form provides real-time feedback on user input. A red border and error message appear for the 'E-mail address' field, indicating it is not in a valid format. A green checkmark and message confirm that the 'Password' is good.
- Sequencing:** The fields are sequenced in a logical order, starting with personal information, followed by email and password, and then security questions.
- Visual Appeal:** The form uses clear labels, asterisks to denote required fields, and distinct colors for error and success messages to enhance its visual clarity and user experience.

Form Fill-In

- Elements of Form Fill-In UID
 - Familiar field labels
 - “Home address” instead of “domicile”
 - Consistent terminology
 - Consistent abbreviations
 - Visible space and boundaries for data-entry fields
 - Required fields clearly marked
 - Privacy information
 - Users will be anxious sharing their personal information and wonder how the data will be used and by who
 - Completion signal
 - Explicit submission and confirmation

Create an IEEE Account ?

*** Required field**

Provide your personal information

*** Given/First name:**
Catherine

Middle name:

*** Last/Family/Surname:**
Plaisant

Enter e-mail address & password
The e-mail provided here will be the username of your account.

*** E-mail address:**
cplaisant@

*** Re-enter e-mail address:**

*** Password:**

*** Confirm password:**

The e-mail address provided is not in a valid e-mail format (for example: j.doe@nomail.com). Please try again.

Your password is good
Passwords must be between 8 and 64 characters, and include at least one number. [More...](#)

Set security questions
For your security, IEEE Accounts are required to have two security questions and answers.

*** Security question 1:**
Select

*** Type your answer:**

*** Security question 2:**
Select

*** Type your answer:**

[Privacy & Opting Out of Cookies](#)

Create Account and Continue Joining [Cancel](#)

Form Fill-In

- Format-Specific Fields

- Using custom widgets and direct manipulation can facilitate data entry
 - Calendars can be used to enter dates
 - Seating maps can help users select airplane seats

- Telephone numbers

- Dates

- Times

- Money amounts

- Passwords

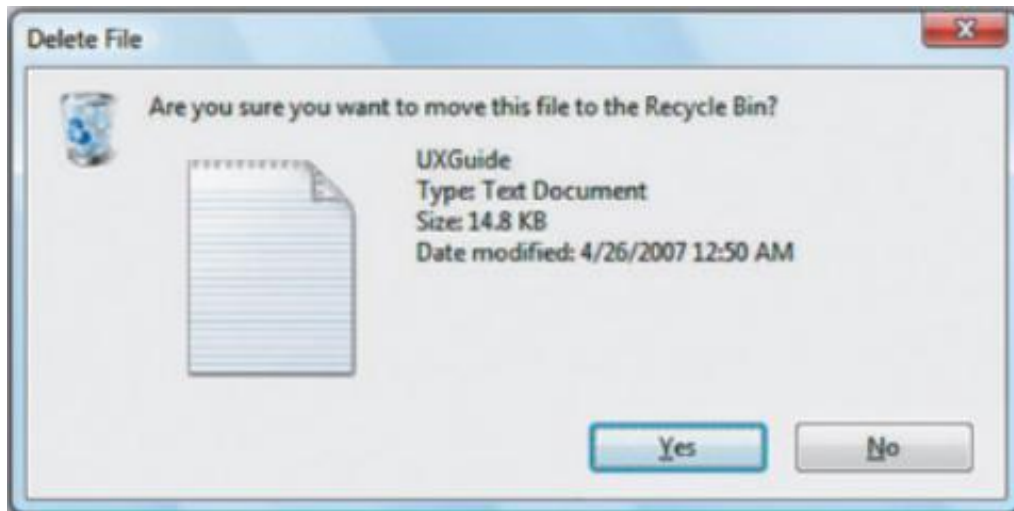
- Two-factor authentication (2FA) requires you to provide two distinct forms of identification to access an account

Guidelines for password creation

- Use two-factor authentication for secure accounts.
- Indicate the rules for password creation.
- Ask for the password to be entered a second time.
- Hide the password with **** by default for privacy.
- Provide an option to unhide the password.
- Provide feedback encouraging strong password selection.

Dialog Boxes

- Conception
 - Many tasks are interrupted to request users to select options, perform limited data entry, or review alerts and error messages
 - The most common solution is to provide a dialog box



Signal Word

Provide clear visual cues and type of alerts

Nature of hazard

Provide succinct reason for the alert

Actions

Provide a list of actions to respond to the alert

User Feedback

Provide ability to capture user feedback

A screenshot of a 'Drug - Drug Interaction' warning dialog box. The title bar is not visible. The main content area has a red warning triangle icon followed by the text 'WARNING! Drug - Drug Interaction'. Below this, the text 'Warfarin - Aspirin' is displayed in bold, followed by 'Increased risk of bleeding' and a link to 'guidelines'. The section is titled 'Management'. There are three buttons: 'Aspirin' (Keep Aspirin, do not order Warfarin), 'Warfarin' (Keep Warfarin, cancel Aspirin), and 'Override' (Order both Warfarin and Aspirin). The 'Override' button is highlighted in red. To the right of the 'Override' button is a checkbox labeled 'Confirm override'. Below the 'Override' button, there is a note: 'Check INR frequently and advise patient for warning signs of bleeding'. At the bottom right, there is a 'Cancel' button. A link 'Provide feedback on this alert' is located at the bottom left of the dialog box.